IT-520 Final Project

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Background and Area of Focus

BoldFlash is an organization founded with the intention of designing and manufacturing computer storage media, initially designing floppy disks and more recently focusing on solid-state drives (SSDs) and products that utilize them, such as MP3 players, USB flash drives, and laptop computers (Beer & Shelton, 2012, p. 2). In January of 2012, Dr. Roger Cahill was promoted to Vice President of BoldFlash's Mobile division, discovering that major shifts in the company's recent history such as rapid sales growth, corporate restructuring, and fast project development had hindered communication and in turn reduced the efficiency of cross-division performance, negatively affecting the organization's supply chain (Beer & Shelton, 2012, p. 2).

In response to the difficulties introduced by poor communication, Dr. Cahill has appointed me as the director of technical communications of the mobile division, tasked with creating and implementing communication guidelines to improve correspondence between Boldflash's internal departments, its associates, and customers. The introduction of these standards should also serve to reduce the amount of communication necessary to convey ideas, improve supply chain function, and mitigate issues caused by organizational growth.

To begin this initiative, divisions associated with product documentation will be the first to receive defined communication guidelines. As noted in an email provided by a Technical Support team manager, installation instructions associated with products such as the A23-B flash drive are rife with spelling errors or otherwise offer incomplete information. For example, the A23-B flash drive's installation instructions utilize inconsistent directory naming conventions and vocabulary, poor filesystem hierarchy visualization, nonspecific software installation instructions, and the absence of information necessary for customers to contact the BoldFlash technical support team.

Successful adherence to the implemented communication standards will provide higher quality instructions that assist customers with product installation to a greater degree and reduce the technical support team's workload.

Key Stakeholders

The first key stakeholder to address (of the divisions associated with product documentation) is Research & Development (R&D). Serving as a linchpin within BoldFlash, R&D is solely responsible for creating new products, improving existing products, and providing documentation regarding those products and product adjustments. Subpar documentation and poor communication regarding product specifications may burden other divisions within BoldFlash. As outlined in the example provided by the Technical Support team's manager, spelling and grammar errors, inconsistent naming conventions, and other issues serve to cause problems within Technical Support. Improving documentation standards and internal correspondence should alleviate these issues.

Other key stakeholders affected by product documentation are Sales and Marketing, Technical Support, and customers seeking to buy the product. Receiving well-formatted, correct, and thorough documentation serves to assist the Sales team in creating detailed and accurate marketing materials, in turn allowing the team to sell the product more effectively. Similarly, Technical Support is dependent on specifications and comprehensive instructions to assist customers with any issues they may have installing or using new products. By improving the quality of correspondence from R&D, the customer in turn benefits from greater knowledge of products that may fit their needs, and prepared technical support should they need assistance.

Established Practices

To improve BoldFlash's internal communication, various strategies should be considered for clarity and technical accuracy; the installation instruction document provided by the Technical Support team is an ideal example of how communication practices can be implemented.

The first strategy to consider is proofreading. The instruction document contains several typos, punctuation errors, and other grammatical errors that may confuse customers and technical support specialists such as unclosed parenthesis and the inconsistent application of end-of-sentence punctuation. Most notably, inconsistent naming conventions may be cause of the most confusion; directories are referred to as "dir" and "subdirectory" interchangeably without clarifying that they are synonymous. The Microsoft Style Writing Style guide suggests that technical terms should only be used when they are the clearest way to communicate a concept and should be used consistently (Microsoft, 2022). Following that standard, a more commonly associated term such as the Windows equivalent "folder" may be an acceptable alternative to "directory," assuming implementing repeated use of "directory" is unsatisfactory for the document's tone.

To provide further clarity to the instruction document, the use of spatial organization when describing filesystem hierarchies will provide greater clarity than listing a series of directories and a loose description of their files. Indentation and bullet points may be sufficient to quickly provide a visual example of a filesystem, but use of graphics in addition to text depicting how folders and files are stored relative to the main directory will provide a stronger example and reduce confusion for Technical Support and customers. (Gerson & Gerson, 2016, pp. 69, 162).

Use of specific details when communicating instructions should be liberally utilized to clarify the installation process. While writing phrases like “click on OK” may reduce the time needed to write instructions, if a program throws an error at a specific point during installation, Technical Support will benefit from additional information to troubleshoot a customer’s issue, for example (Gerson & Gerson, 2016, pp. 56-57). Referring to specific installation messages or providing screenshots of the application steps provides readers and Technical Support with a frame of reference to further assist with problem solving. Including specific information as to how to contact Technical Support such as customer support phone numbers, email addresses, or URLs sending customers to a support form may serve to simplify the support process. Clear instructions and product details will also allow the Sales team to make accurate marketing materials and direct those materials to different audiences depending on the device’s capabilities and potential place in a personal or professional space.

Implementing these practices should serve to improve communication between departments, bolster sales with clear advertising, and ultimately assist customers and deliver a higher quality product overall.

Training Plan: Communication

To begin implementing these training strategies, capitalizing on “downward” communication may be imperative to reach as many members of the organization as quickly as possible. As outlined in the University of Minnesota’s documentation on Human Resource Management, focusing on training management in the new communication team’s standards with the intent of distributing that information downward to their subordinates, who in turn continue to inform and train in these new standards to the base levels of the organization (University of Minnesota, 2016, p. 9.1). Communication flowing upward and horizontally should similarly be

utilized to facilitate an environment where asking questions and offering suggestions to team members or senior members of staff is encouraged to reduce confusion and further streamline the communication standards set; surveys may also be helpful for streamlining the process of receiving feedback and improving communication guidelines (University of Minnesota, 2016, p. 9.1).

In addition to encouraging communication between levels of the organization to ensure the standards set are being understood, strategies such as active listening should be utilized. Active listening is defined as the process of listening with the intention of not only hearing and processing what the speaker is saying, but also verifying what the speaker is saying for accuracy (University of Minnesota, 2016, p. 9.1). This is directly contrasted with combative listening, where the listener is focused on pushing a specific point of view as opposed to listening with the intent of learning, and passive listening, where the listener is listening, but not verifying the information they have received (University of Minnesota, 2016, p. 9.1). While passive listening may seem like an effective way to take in information, mishearing or unclearly written correspondence may lead to confusion and reduce productivity within different branches of the organization.

In every aspect of the new communication guidelines to be implemented, useful feedback will be necessary to ensure that correspondence within BoldFlash is effective and succinct, reducing the time needed to communicate overall and improving the organization's ongoing performance.

Training Plan: Collaboration

Creating an effective training curriculum and communications team should focus on incorporating members from as many aspects of the organization as possible. Every department

within BoldFlash will operate differently, in turn facilitating the need for different requirements in communication standards (Gerson & Gerson, 2016, pp. 8-9). Recruiting members from a variety of departments and encouraging collaboration should serve to implement standards and training procedures that serve every aspect of the organization as opposed to a small assortment of supposed "one size fits all" solutions that may cause confusion or neglect to format technical information adequately (Gerson & Gerson, 2016, pp. 8-9).

To foster collaboration, implementing enterprise social networks (ESNs) may encourage knowledge sharing between members of the organization not only between the curated members of the communications team, but throughout the whole organization (Biech, 2014, pp. 89-92). Additionally, adopting ESNs may offer avenues for training that are not typically accessible through standard training workshops and methods taught to management; planning online events that encourage social activity within the organization while also providing educational resources to foster communication may be more effective in engaging and ultimately preparing BoldFlash employees for disruptive changes in the current communication systems (Biech, 2014, pp. 94-95). Utilizing electronic training methods in addition to traditional in-person methods may also serve to assist BoldFlash's international divisions; as such, easily translated material should be considered and applied whenever possible.

These approaches should serve to provide R&D, Sales, and Technical Support (in addition to the rest of the organization) with the communication skills necessary to identify what composes useful marketing material, in turn improving the overall customer experience.

Training Plan: Methods

For initial training between the communications team and department management, methods such as international assignment training and web-based training may be effective solutions for the global nature of BoldFlash's operations. International assignment training may serve to bridge cultural divides between branches in different nationalities, educating staff on different social norms, cultural similarities, and language skills that may strengthen relationships between staff members in different countries and mitigate the risk of accidentally offending any parties involved (University of Minnesota, 2016, p. 8.3). To build skills relevant to technically communicating ideas, curated web-based training may be sufficient for providing several learning methods to capture as many learning styles as possible (Ledford & Sleeman, 2002, pp. 189-190). Web-based training can provide an interactive outlet that offers visual and auditory examples of the material being implemented in addition to actively testing management to ensure the concepts being conveyed are understood (University of Minnesota, 2016, p. 8.3).

After senior staff has been prepared for the new communication guidelines, training can then be conveyed to subordinates through a combination of the aforementioned electronic learning platform and group discussion or a brown bag meeting to further ensure that the new communication standards are understood (Biech, 2014, pp. 150-151). This combination utilized in tandem with vertical and horizontal communication should be sufficient for preparing staff for effectively communicating the ideas necessary to improve documentation. Additionally, given the nature of BoldFlash and the departments that will be initially targeted by this communication initiative, the technical knowledge associated with the documentation and subsequent marketing should be sufficient to facilitate effective computer training without the setbacks that may occur at a less technically-inclined organization (Ledford & Sleeman, 2002, pp. 188-189).

Framework for Communication

After specific guidelines have been outlined and staff have been trained, different communication methods will be utilized to capitalize on their effectiveness for various tasks. Memorandums should be utilized to announce upcoming major product changes or changes in operating procedure such as the implementation of new communication guidelines (Gerson & Gerson, 2016, p. 186). In the event more detailed correspondence is necessary within the organization, e-mails containing questions, answers, or attached schematics (and other technical documents) should be used (Gerson & Gerson, 2016, p. 186). These methods should provide a solid foundation for the majority of BoldFlash's technical communication, allowing for a greater amount of detail and targeted messaging to ensure information is being transferred to allow the remainder of the organization to make the most informed decision at any given time.

For more succinct communication, instant messaging platforms (that may be included and incorporated within an ESN) may be sufficient for answering short questions and building employee camaraderie, further bolstering organizational cohesion and encouraging collaboration (Gerson & Gerson, 2016, p. 186). For the Sales department to effectively communicate with the BoldFlash customer base, striking branding may be considered in addition to detailed product specifications and continued communication from R&D to bolster sales performance and improve the BoldFlash brand image (Goldfarb & Tucker, 2015). While advertising standards are slightly outside of the scope of this document, they may be necessary for BoldFlash's ongoing success and building a stronger customer base as the organization continues to grow.

Target Audience

Before correspondence is drafted, the author should consider the target audience of that document. Different target audiences have different needs and may not share the same level of experience, placing greater importance on choosing what information is imperative to share with that audience. A prospective customer base may not be aware of write speeds, encryption methods that a new BoldFlash Flash Drive may include, or minutiae of the controller's operation; as such, communication should accommodate and provide appropriate context when necessary. In this case, the documentation provided by R&D should reflect the needs of two targeted departments and, peripherally, one external audience: Sales and Marketing, Technical Support, and BoldFlash customers.

While the primary focus of R&D's Flash Drive installation instructions document is to assist internal departments with creating advertising materials and training documentation, the BoldFlash customer base will be either directly or indirectly affected by the quality of the R&D department's work; if the provided documentation is unclear, customers may be reluctant to purchase the new product or require more assistance operating the product, causing BoldFlash to incur greater costs in either case.

To best revise the Flash Drive installation instructions document, the audience's preexisting knowledge of Flash Drive operation, their informational needs, and their relationship to the R&D department needs to be considered to best tailor the core message to those needs (Texas A&M University, 2010). Given that the primary target audiences are the Sales and Marketing and Technical Support departments with the intention of providing the information necessary to ultimately assist customers, a degree of competency can be assumed as each department exists within BoldFlash. Assuming that these departments are already familiar with

most of BoldFlash's products, the R&D department should emphasize the new features the Flash Drives offer that may be unfamiliar and cause confusion. Finally, while the Sales and Marketing and Technical Support teams may be aspects of the BoldFlash organization, the global nature of BoldFlash and its partners should be considered when choosing an appropriate tone for this correspondence; a professional and informational tone may best serve the purpose of this documentation and improve the quality of information overall.

Message

The intended message of this message is to provide installation instructions for the BoldFlash Flash Drive #A23-B product, assisting the Sales and Marketing and Technical Support teams with their respective roles providing marketing materials for the product and assisting customers after the product has been released. This is pertinent to the Sales and Marketing team as the details contained within this document and documents like it will be used to determine possible selling points and to identify new product features for the new Flash Drives and other products. Similarly, the Technical Support department will use these documents to create appropriate training material for their staff to assist customers with the installation and operation of these new products. Identifying how these Flash Drives operate will also provide insight into potential breaking points in the product's operation such as incompatible driver updates, further assisting the Technical Support team with diagnosing issues customers may be having.

By providing clear and detailed installation and operation instructions for the #A23-B Flash Drives, BoldFlash stands to benefit from important context that will improve the quality of marketing and training materials, assisting customers, and boosting sales of the new Flash Drives.

Critique

The BoldFlash Flash Drive instructions utilize a chronological layout to assist BoldFlash employees with installation, starting with the process of preparing the Flash Drives with driver installation, providing an example of the intended filesystem structure for operation, and concluding with first time operation instructions. While the general structure is useful, the content of each section contains errors with inconsistent terminology, spelling errors, or unprofessional tone. These errors and design decisions may serve to confuse readers, reducing the productivity of the Sales and Marketing and Technical Support teams in addition to potentially causing issues for customers if not addressed in time.

Starting with the first section, a base directory is created and named "FlashDrive;" the "directory" nomenclature is appropriate for macOS filesystems, but there is no explicit acknowledgement that these instructions are intended for macOS computers. Given that Linux filesystems similarly use directories as opposed to Windows' use of "folders," it is important to address the intended system for these installation instructions. In the next step, the document erroneously refers to the "FlashDrive" directory as "FlashDriver" and suggests creating a subdirectory within called "Driver." While a simple mistake, the addition of another letter may be a source of confusion, implying that the #A23-B Flash Drive may include directories that do not actually exist at the time of driver installation.

The next few steps address installing drivers and file requirements from a BoldFlash repository without specifying what files and directories are necessary for the product's operation, creating ambiguity and possibly creating another source of confusion within the document. The reader is then instructed to download the Irrlicht 1.8 and Irrklang 1.4.0 Mac drivers from their respective URLs and install them within a nebulous "main" directory, introducing a second term

for the “base” directory here named “FlashDrive” and another potential source of unclear and confusing communication. Given the repetitive nature of these steps, consolidating them under a single section (e.g. “Download and extract the following compressed files into the ‘/FlashDrive/’ directory) may create a clearer task list and avoid redundant language use.

Throughout the “prerequisites” section, the term “directory” is exchanged for the simpler “dir” without addressing what “dir” is in reference to. This may be a potential cause of confusion, creating ambiguity around the term and whether directories are related to “dir” or a separate concept altogether (Team Acolad, 2022).

The second section does not follow the title schema established by the first section but provides a loose description of the intended filesystem structure. The FlashDrive and Driver directories are listed in their creation order but would benefit from indentation to establish their hierarchy as the Driver directory and the .cpp and .h files contained within are inside of the FlashDrive directory. The document then details that two Irrlicht directories, a lib directory, and the irrKlang directory should be within the filesystem; indentation should be used here to identify that these directories are inside of the FlashDrive folder, with further indentation used to provide a visual reference of what files are contained within each directory. This section would also benefit greatly from improved punctuation, sentence structure, and a more professional tone.

The final section provides guidelines for product installation, following a chronological order like that of the first section, beginning with inserting the product into a USB port. The next two steps outline a nebulous process of advancing through installation dialogue boxes by clicking the “OK” button with no specific detail as to what sections may require greater user involvement, such as what elective software may be added or specific steps that may have increased risks of causing errors for users. By including more detailed first-time installation

instructions, Technical Support would have more context for identifying issues users may have during the installation process. The final step then enthusiastically states that when a message states the software has been successfully installed, the Flash Drive is ready to be used. Readers are then told to contact Technical Support should they have issues without providing the appropriate channels to contact the department. This section would also stand to benefit from more professional language choice and more consistent spelling and punctuation.

By addressing spelling errors, adjusting for a more professional tone, and utilizing technical jargon consistently, R&D and other departments within BoldFlash stand to benefit from straightforward and clear communication that benefits not only other branches within the organization, but the customers and distributors that communication may indirectly affect.

The next section of this document will provide an example of a revised version of the R&D department's #A23-B training document and a brief explanation of the changes included.

Part Two

Revised Installation Instructions

BOLDFlash Flash Drive Product #A23-B

Mac (macOS) Installation Instructions

1. Prerequisites
    - Create a base directory (here referred to as 'FlashDrive')
    - Create a subdirectory inside of 'FlashDrive' (here referred to as 'Driver')
    - Download all files and directories within the A23-B file repository
    - Place the A23-B repository files in the 'Driver' subdirectory
    - Download the following files and extract them into the base 'FlashDrive' directory
        - Irrlicht 1.8
            - (http://irrlicht.techtoys.net/downloads.html)
        - Irrklang 1.4.0 Mac/macOS Driver
            - (http://www.techtoys.com/irrklang/downloads.html)
2. File System

    The resulting structure should resemble the following, with each level of indentation representing a deeper level in the file system's hierarchy:
    - FlashDrive base Directory
        - Driver subdirectory
            - A23-B repository files
        - lib subdirectory
            - lib files
        - IrrKlang 1.3.0 subdirectory
            - irrKlang files
        - Irrlicht subdirectory
            - Irrlicht .cpp and .h files
        - Irrlicht 1.8 subdirectory
            - Irrlicht curl files
3. First Time Operation:
    - Insert the #A23-B Flash Drive into an open USB port
    - An installation prompt to install the #A23-B Flash Drive drivers should appear
    - Progress through the installation as necessary
        - (Greater detail in this section should be provided by Research and Development)
    - When installation is completed, please enjoy your new #A23-B Flash Drive!

Problems with installation?

Please contact our Technical Support Team at: ###-###-#### (phone number needed)

Or e-mail us at: techsupport@boldflash.com (e-mail address needed)

Explanation

The revised BoldFlash #A23-B Flash Drive installation instructions begin by specifying the product's number as in the original document, but then specifying the intended system for these instructions. Given that Windows utilizes a "folder" filesystem schema and Linux shares the "directory" filesystem schema, specifying the intended system for this document should serve to reduce confusion that may be caused by incompatible drivers or software exclusive to macOS "Mac" systems.

The first section now uses consistent terminology for the base directory and its subdirectory, 'FlashDrive' and 'Driver,' respectively, and references to their level of hierarchy to establish how the directories are arranged. Next, a specific repository containing the necessary files for #A23-B is named, providing a single location that contains only the files necessary for running the Flash Drive, reducing the risk of downloading irrelevant files that may cause performance issues. The files are then placed within the 'Driver' subdirectory. Finally, the reader is instructed to download and extract the Irrlicth 1.8 and Irrklang 1.4.0 Mac drivers from their respective URLS to the base 'FlashDrive' directory, reducing redundancy but still providing specific information for installation.

The next section, now labelled "File System," outlines the intended directory structure with a brief explanation of how the directory should be read; while specific information as to what files reside in which subdirectories is unavailable, the use of indentation should provide a better visual example of how the filesystem is organized.

The final section, now called "First Time Operation," provides a nonspecific overview of how the Flash Drive should be installed on a new system with revised and more professional

language. The BoldFlash R&D department should be contacted to provide more specific instructions regarding additional installation instructions, such as designating a directory to install additional software or choosing an encryption type should it be necessary. For the final step, the enthusiastic tone of the original document has been replicated to provide some levity in this otherwise very technical document.

Finally, the Technical Support section has been revised to include an invitation to contact BoldFlash should a user have any problems, now including sections for the Technical Support phone number and e-mail address to make the process of contacting customer support as easy as possible.

By updating the language of this document to address internal departments but still appeal to BoldFlash customers and providing specific details regarding the product's preparation and installation, this documentation should better serve the needs of the Sales and Marketing and Technical Support departments by reducing confusion and providing a more streamlined process overall. Implementing similar changes to documentation moving forward should provide greater clarity and reduce the time needed to successfully communicate between departments, improving the efficiency of BoldFlash as an organization entirely.

Future Growth and Plan Adaptation

As BoldFlash solidifies its communication processes, the standards demonstrated in this document can and should be adjusted to meet the needs of other pieces of documentation, such as the provided BoldFlash Technical Service Communication Artifact or the Internal Business Process Artifacts.

The first portion of the BoldFlash Technical Service Communication document may benefit from additional context. While it can be assumed that Sales and Technical Support have a degree of familiarity with the technical specifications of BoldFlash products, R&D providing these teams with explanations of less intuitive specifications or leaving out less imperative specifications may serve to streamline the process of creating new marketing materials without slowing the Sales team down with unnecessary and confusing jargon. While a few term definitions are defined on the next page, these definitions also lean jargon-heavy, with undefined abbreviations (such as FPDMA, LBA, and MTBF) providing possible sources of confusion within the document. Simple definitions to these terms and a single brief explanation of each abbreviation's meaning should serve to greatly improve this document's communicative efficacy.

The Internal Business Process Artifacts seem to juxtapose the Technical Service Communication document, providing examples of communication that aren't detailed enough. Sending documents to "Kevin's admin assistant" is not sufficient for readers to contact this person should they need to; providing more specific information could assist the department in keeping sensitive information secure and quickly contacting the correct person to receive answers to questions they may have. Similarly, removing product documentation "sitting on a server" makes the information more difficult to find and providing a specific location on the fileserver could improve the time needed to fetch that information.

In the second paper, a vague idea of problems is referred to regarding customer orders missing their requested delivery dates. Providing specific details as to the sources and frequency of those problems could greatly assist with identifying an underlying cause, allowing for BoldFlash teams to take the appropriate action as opposed to assuming the problem lies with

Manufacturing. The final paper suffers from a similar problem, referring to nebulous ideas of "feedback" from an unidentified source that could provide more context as to the specific source of the problem; by introducing more specific language in this case, BoldFlash stands to benefit from improved communication and more easily identified problems.

Technical communication is a delicate balance, but these standards should serve to direct the organization to an ideal middle ground, providing an adequate amount of information necessary for success without inundating readers with excessive detail.

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